

Problem 4

a) Solved code.

FILE: ~~resultss.csv~~ results.txt TR.csv
TR.txt ~~renamed~~ ✓

b) For a tabula recta to be correct we have to have a permutation of the alphabet on each line and on each column. We don't want to have duplicates (especially on columns, because decryption would not be possible in this case).

YES, it is correct because we ~~do~~ have a perm. of the alphabet on lines & cols. Also a, b are coprime with N .

c) $a, b \rightarrow$ coprime with $m \Rightarrow$ we will have perm. on rows and cols.

~~Let's assume a, b are not coprime with m~~
 $a, b < m$, obvious, we work in $\mathbb{Z}_m$
 $c < m \rightarrow$ works like a constant shift.

d) $V(a, b, c)$

As said a, b - coprime with m

$$m = 26 = 2 \cdot 13 = p \cdot q$$

Number of values coprime with 26: $p^{q-1} - \underbrace{(q-1)}_{\text{nb. of multiples of } p} - \underbrace{(p-1)}_{\text{nb. of multiples of } q}$

$$= p^{q-1} - q + 1 - p + 1$$

$$= p(q-1) - (q-1)$$

$$= (q-1)(p-1)$$

$$= 12 \cdot 1 = 12$$

$\Rightarrow$ "a" can take 12 values
"b" can take 12 values

"c" can take 26 values $\rightarrow$ only condition $c < m$

$$\Rightarrow 12 \cdot 12 \cdot 26 = 3744 \text{ versions of } V(a, b, c)$$

Problem 4 - continued

e) YES it is possible.

Because the algorithm works like Vigenere, it repeats the key so that the keystream to be as long as the message.

By analysing the repeating groups of letters in the ciphertext and determining the possible key length, the cryptogram can be put into columns and on each column freq. analysis.

In conclusion, Kasiski method works on the $V(a,b,c)$ encryption.